```
abstract class EnergySource {
    String sourceId;
    String sourceName;
    double energyGenerated;

    EnergySource(String sourceId, String sourceName, double
energyGenerated) {
        this.sourceId = sourceId;
        this.sourceName = sourceName;
        this.energyGenerated = energyGenerated;
    }

    abstract double calculateEfficiency();

    void displayDetails() {
        System.out.println("Source ID: " + sourceId);
        System.out.println("Source Name: " + sourceName);
        System.out.println("Energy Generated: " + energyGenerated + "
kWh");
        System.out.printf("Efficiency: %.2f%%\n", calculateEfficiency());
        System.out.println("-----------------------------------");
    }
}

class SolarEnergy extends EnergySource {
    SolarEnergy(String sourceId, String sourceName, double energyGenerated)
{
        super(sourceId, sourceName, energyGenerated);
    }

    @Override
    double calculateEfficiency() {
        return (energyGenerated / 5000.0) * 100.0;
    }
}

class WindEnergy extends EnergySource {
    WindEnergy(String sourceId, String sourceName, double energyGenerated)
{
        super(sourceId, sourceName, energyGenerated);
    }

    @Override
    double calculateEfficiency() {
        return (energyGenerated / 8000.0) * 100.0;
    }
}

public class EnergyManagementSystem {
    public static void main(String[] args) {
        EnergySource ref;

        ref = new SolarEnergy("SOL-101", "Sahara Solar Park", 3750.0);
        ref.displayDetails();

        ref = new WindEnergy("WND-202", "Alta Wind Energy Center", 5600.0);
```

```
        ref.displayDetails();
        System.out.println("Monish K A, 1BF25CS157");

    }
}
```

```
Source ID: SOL-101
Source Name: Sahara Solar Park
Energy Generated: 3750.0 kWh
Efficiency: 75.00%
---------------------------------
Source ID: WND-202
Source Name: Alta Wind Energy Center
Energy Generated: 5600.0 kWh
Efficiency: 70.00%
---------------------------------
Monish K A, 1BF25CS157
```

```
interface ParkingCharge {
    double calculateCharge();
    void displayParkingDetails();
}

class VehicleParking {
    String vehicleNumber;
    String vehicleType;
    int parkingHours;

    public VehicleParking(String vehicleNumber, String vehicleType, int
parkingHours) {
        this.vehicleNumber = vehicleNumber;
        this.vehicleType = vehicleType;
        this.parkingHours = parkingHours;
    }
}

class TwoWheeler extends VehicleParking implements ParkingCharge {

    public TwoWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Two Wheeler", parkingHours);
    }

   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 20.0;
        } else {
            return (2 * 20.0) + ((parkingHours - 2) * 10.0);
        }
    }

   public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " +
calculateCharge());
```

```
        System.out.println("-----------------------------------");
    }
}

class FourWheeler extends VehicleParking implements ParkingCharge {

    public FourWheeler(String vehicleNumber, int parkingHours) {
        super(vehicleNumber, "Four Wheeler", parkingHours);
    }

   public double calculateCharge() {
        if (parkingHours <= 2) {
            return parkingHours * 50.0;
        } else {
            return (2 * 50.0) + ((parkingHours - 2) * 30.0);
        }
    }

    public void displayParkingDetails() {
        System.out.println("Vehicle Number: " + vehicleNumber);
        System.out.println("Vehicle Type: " + vehicleType);
        System.out.println("Parking Hours: " + parkingHours + " hrs");
        System.out.println("Total Parking Charge: Rs. " +
calculateCharge());
        System.out.println("-----------------------------------");
    }
}

public class ParkingManagementSystem {
    public static void main(String[] args) {

        ParkingCharge parkingRef;

        parkingRef = new TwoWheeler("KA-01-AB-1234", 5);
        parkingRef.displayParkingDetails();

        parkingRef = new FourWheeler("KA-05-CD-5678", 4);
        parkingRef.displayParkingDetails();
        System.out.println("MONISH K A,1BF25CS157");

    }
}
```

```
Vehicle Number: KA-01-AB-1234
Vehicle Type: Two Wheeler
Parking Hours: 5 hrs
Total Parking Charge: Rs. 70.0
-----------------------------------
Vehicle Number: KA-05-CD-5678
Vehicle Type: Four Wheeler
Parking Hours: 4 hrs
Total Parking Charge: Rs. 160.0
-----------------------------------
MONISH K A,1BF25CS157
```